Upward Propagation of Nonlinear Abel Averages

A partial result for the open question in arXiv:1311.6698

Automated mathematics run `fa_banach_001`, lane 8

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Status. Candidate partial result, likely valid; human review requested. The original range below the known parameter remains open.

1 Source problem

Bracci, Kozitsky, and Shoikhet study Abel averages of nonlinear maps on the open unit ball $\mathbb{B}$ of a complex Banach space X [1]. In the case $\omega = 0$, the Abel average at a positive parameter α is

$$\Phi_\alpha = (\mathrm{Id} - \alpha h)^{-1}.$$

Their final question asks whether, for a balanced dense set $D \subset \mathbb{B}$ and a weakly closed, 0-dissipative map $h : D \to X$, existence of one holomorphic Abel average $\Phi_{\alpha_0} : \mathbb{B} \to D$ implies existence for every $\alpha > 0$.

$\mathrm{Null}(h) = \{(\xi, \eta) \in \mathbb{B}^2 : \lambda\xi = -\epsilon\eta^2\}$, which would imply that $\mathrm{Null}(h)$ be a one-dimensional holomorphic retract of $\mathbb{B}^2$ (so-called *complex geodesic of* $\mathbb{B}^2$), while one-dimensional holomorphic retracts of $\mathbb{B}^2$ are known to be just the intersection of affine complex lines with $\mathbb{B}^2$.

4. Open question

Let D be a balanced dense subset of B, and let h be a closed (weakly closed) map on D with values in X. Assume that h is 0-dissipative and for some $\alpha_0 > 0$ its Abel average exists, is holomorphic, and maps B into itself. Do the Abel averages exist for all positive α?

The image is a full-width crop of page 13 of the source PDF, Section 4, "Open question."

2 Proof intuition

The nonlinear resolvent identity can be arranged so that larger parameters are obtained from a fixed point problem involving only the known resolvent. If μ is the known parameter and $\alpha \geq \mu$, put $t = \mu/\alpha$. For fixed $x \in \mathbb{B}$, consider

$$K_x(z) = tx + (1-t)\Phi_\mu(z).$$

This is a convex combination of the fixed interior anchor x and a point of $\mathbb{B}$. Consequently its whole image stays a positive distance from the boundary of $\mathbb{B}$. The Earle–Hamilton theorem gives a unique fixed point $Z_t(x)$, holomorphic in x. Applying Φ_μ to this point produces the desired larger-parameter resolvent. Convexity also supplies the key uniqueness argument: any competing solution at parameter α generates a point of $\mathbb{B}$ and hence must pass through the same fixed point.

This mechanism reverses for $\alpha < \mu$: the affine identity becomes an extrapolation rather than a convex combination. That is precisely the part of the source question not settled here.

3 Upward propagation theorem

We use "well defined" in the natural inverse sense: for every $w \in \mathbb{B}$ there is a unique $u \in D$ satisfying $u - \mu h(u) = w$.

Theorem 1 (Upward propagation)**.** *Let X be a complex Banach space with open unit ball $\mathbb{B}$, let $D \subset \mathbb{B}$, and let $h : D \to X$. Suppose that for some $\mu > 0$,*

$$\Phi_\mu = (\mathrm{Id} - \mu h)^{-1} : \mathbb{B} \longrightarrow D$$

is well defined and holomorphic. Then for every $\alpha \geq \mu$ the Abel average

$$\Phi_\alpha = (\mathrm{Id} - \alpha h)^{-1} : \mathbb{B} \longrightarrow D$$

is well defined and holomorphic.

More precisely, with $t = \mu/\alpha$, let $Z_t(x)$ denote the unique fixed point in $\mathbb{B}$ of

$$K_x(z) = tx + (1-t)\Phi_\mu(z).$$

Then

$$\boxed{\Phi_\alpha(x) = \Phi_\mu(Z_t(x))}.$$

Proof. Fix $\alpha \geq \mu$ and write $t = \mu/\alpha \in (0, 1]$. For $x \in \mathbb{B}$, the map $K_x : \mathbb{B} \to \mathbb{B}$ is holomorphic. It maps $\mathbb{B}$ strictly inside $\mathbb{B}$. Indeed, set $\delta = 1 - \|x\| > 0$. If $z \in \mathbb{B}$ and

$$\|w - K_x(z)\| < t\delta,$$

write $e = w - K_x(z)$. Then $x + e/t \in \mathbb{B}$, and

$$w = t\left(x + \frac{e}{t}\right) + (1-t)\Phi_\mu(z) \in \mathbb{B}$$

by convexity. Thus the ball of radius $t\delta$ about every point of $K_x(\mathbb{B})$ lies in $\mathbb{B}$.

The Earle–Hamilton fixed-point theorem [3, Theorem 2] now gives a unique fixed point $Z_t(x) \in \mathbb{B}$. Moreover, $x \mapsto Z_t(x)$ is holomorphic. For completeness, on any neighborhood on which $\|x\| \leq r < 1$, the strict-inside margin above is uniformly at least $t(1-r)$. The Earle–Hamilton contraction argument therefore makes the iterates of K_x locally uniformly Cauchy in x; their holomorphic limit is $Z_t(x)$. This is the same parameter-dependence argument used in the proof of [3, Theorem 3].

Set $z = Z_t(x)$ and $y = \Phi_\mu(z) \in D$. Since z is fixed by K_x,

$$z = tx + (1-t)y.$$

On the other hand, the definition of Φ_μ gives $z = y - \mu h(y)$. Comparing these identities yields

$$t(x - y) = -\mu h(y), \qquad \text{hence} \qquad x = y - \frac{\mu}{t}h(y) = y - \alpha h(y).$$

Therefore a solution exists, it belongs to D, and the displayed formula for Φ_α is holomorphic.

It remains to prove that the inverse is single valued on $\mathbb{B}$. Suppose $\widetilde{y} \in D$ also satisfies $x = \widetilde{y} - \alpha h(\widetilde{y})$. Define

$$\widetilde{z} = \widetilde{y} - \mu h(\widetilde{y}) = tx + (1-t)\widetilde{y}.$$

Because $x, \widetilde{y} \in \mathbb{B}$ and $0 < t \leq 1$, convexity gives $\widetilde{z} \in \mathbb{B}$. The known inverse at μ gives $\Phi_\mu(\widetilde{z}) = \widetilde{y}$, so

$$K_x(\widetilde{z}) = tx + (1-t)\Phi_\mu(\widetilde{z}) = \widetilde{z}.$$

Uniqueness of the fixed point implies $\widetilde{z} = z$, and then $\widetilde{y} = \Phi_\mu(z) = y$. This proves uniqueness and completes the proof. □

Remark 1 (Constructive iteration)**.** *For any starting point $z_0 \in \mathbb{B}$, Earle–Hamilton iteration gives*

$$z_{n+1} = tx + (1-t)\Phi_\mu(z_n) \longrightarrow Z_t(x),$$

locally uniformly in x. Thus the proof supplies an explicit procedure for recovering every larger-parameter Abel average from the one known average.

Remark 2 (Hypotheses not used)**.** *The theorem needs neither balancedness nor density of D, and it needs neither weak closedness nor dissipativity of h. Those assumptions may be relevant only to the unresolved downward direction.*

4 Why the smaller-parameter range remains open

Let $0 < \alpha < \mu$ and put $s = \alpha/\mu \in (0, 1)$. If $z = y - \mu h(y)$ happens to belong to $\mathbb{B}$, the two resolvent equations give

$$x = sz + (1-s)y = sz + (1-s)\Phi_\mu(z).$$

Thus one would need to invert the holomorphic self-map

$$F_s = s\,\mathrm{Id} + (1-s)\Phi_\mu.$$

But a solution at the smaller parameter need not have $y - \mu h(y) \in \mathbb{B}$, and the reverse fixed-point form has a negative affine coefficient. Neither convexity nor Earle–Hamilton controls that extrapolation. The theorem therefore resolves exactly the ray $[\mu, \infty)$ and does not answer the full source question.

5 Verification and novelty bounds

The proof was audited algebraically in both directions. In particular, the strict-inside radius is $t(1 - \|x\|)$, and the uniqueness point $\widetilde{z} = tx + (1-t)\widetilde{y}$ is in $\mathbb{B}$ by convexity. The endpoint $\alpha = \mu$ gives $t = 1$, $Z_t(x) = x$, and recovers Φ_μ.

A bounded literature check searched the exact question text, arXiv id 1311.6698, the source title and authors, "nonlinear resolvent identity," and Earle–Hamilton/resolvent parameter combinations. The 2019 monograph [4] reproduces the question as open. Later resolvent papers found in the search assume a holomorphic generator or a full resolvent family; no explicit answer to the source question or exact statement of the theorem above was found. Because the argument is a short application of a classical fixed-point theorem, novelty confidence is moderate rather than definitive.

Human review should focus on the interpretation of the inverse on $\mathbb{B}$, the standard holomorphic parameter-dependence step, and the uniqueness construction. No computational evidence is used.

References

[1] F. Bracci, Y. Kozitsky, and D. Shoikhet, *Abel averages and holomorphically pseudo-contractive maps in Banach spaces*, J. Math. Anal. Appl. 423 (2015), no. 2, 1580–1593; arXiv:1311.6698. DOI: 10.1016/j.jmaa.2014.10.079.

[2] C. J. Earle and R. S. Hamilton, *A fixed point theorem for holomorphic mappings*, Global Analysis, Proc. Sympos. Pure Math. 16, Amer. Math. Soc. (1970), 61–65.

[3] L. A. Harris, *Fixed points of holomorphic mappings for domains in Banach spaces*, Abstract Appl. Anal. 2003 (2003), no. 5, 261–274. DOI: 10.1155/S1085337503205042.

[4] M. Elin, S. Reich, and D. Shoikhet, *Numerical Range of Holomorphic Mappings and Applications*, Birkhäuser, Cham, 2019. DOI: 10.1007/978-3-030-05020-7.